

Authoritative Data Repository (ADR) Capability Canvas

Business Problem	Fragmented, inaccessible, and poorly managed geospatial data reduces project quality, increases costs, limits data-informed decisions, and constrains the ability to provide GIS as a service.			
Cause(s)	The organization lacks the consistent data, infrastructure, and management practices needed for data-informed decisions. Installation data is not routinely collected or managed, making it difficult for staff and customers to find, trust, and use.			
Goal	- Improve work quality and reduce costs by centralizing, managing, and continually enriching GIS data for project use. - Provide GIS data management and analysis as a customer service. - Maintain a living authoritative repository that supports project selection, planning, and investment decisions.			
Solution Strategy Support project teams and customers through an Authoritative Data Repository for installation GIS data. Assess gaps, improve quality, and expand analytical capabilities as business needs mature.	Business/Process Scope	Data Scope	Technology Scope	People Scope
	Collect, manage, store, and deliver installation GIS data for customer and internal project use.	The repository will contain features and attributes used for planning, design, and the management of projects, facilities, infrastructure, and installations.	An enterprise geodatabase will operate within an approved organizational GIS portal. A central application will provide authorized access to data, web maps, applications, and supporting documents.	Internal teams gain reliable data for better decisions and stronger deliverables. Customers gain managed data and spatial analysis services that support asset, facility, and installation management.
Geospatial Capabilities Collect, manage, store, and deliver installation GIS data as a customer service and as an internal capability supporting project execution.	Describe	Diagnose	Predict	Prescribe
	- Map and compare asset, facility, infrastructure, and utility attributes at specific locations for project, asset, and facility management. - Map and compare facility management statuses. - Map and compare building constraints.	- Determine best locations for facility projects. - Determine where facility management actions are needed. - Determine best route for utility lines. - Determine environmental impacts.	- Forecast future facility needs and where future project locations should be. - Forecast facility management needs. - Forecast utility needs. - Forecast environmental impacts.	- Recommend future projects and their locations. - Recommend future facility management activities. - Recommend utility upgrades. - Recommend mitigation to environmental impacts.
Technology Components Collect, manage, store, and deliver authorized data - Cadastral - Master planning - Facilities - Infrastructure - Utilities - Environmental - Survey - As-built - Elevation - Real property	Data Capture & Integration	Data Analysis & Modeling	Mapping & Data Visualization	Sharing & Communication
	Integrate existing planning datasets first. Add field collection and contractor deliverables through project work. Future collection may incorporate automation, remotely sensed data, and uncrewed systems where appropriate.	Assess existing holdings to identify gaps. Prioritize enrichment activities based on business needs. Potential methods include field collection, remote sensing, survey, and elevation data acquisition.	Present installation information through thematic 2D and 3D web maps selected to match the data format and business use case.	Store data in an enterprise geodatabase. Control viewing and editing through approved groups, roles, and permissions.
Outcomes & Assumptions Reliable and relevant data will support better work and new service opportunities when leadership, funding, customer need, and stakeholder participation remain sufficient.	Expected Outcomes		Design Assumptions	
	Reliable GIS data improves situational awareness, analysis, project quality, and efficiency. Managed data services help customers make better project, facility, and environmental decisions.		- Confirmed demand from internal teams and customers. - Active stakeholder participation and feedback. - Adequate leadership support and funding. - Initial access limited to authorized users.	
Governing Requirements Applicable enterprise geospatial policies require authoritative repositories, consistent data management, standards, controlled access, and stakeholder coordination.	Enterprise Data Policy	Installation Data Policy	Engineering Requirements	Data Management Guidance
	Maintain authoritative geospatial data in an enterprise repository that supports planning, operational decisions, and mission requirements.	Maintain accessible installation data that supports business functions and decision-making across authorized stakeholder groups.	Use centralized and standardized geospatial information to support planning, design, construction, and operations.	Keep data current, validated, secure, and usable. Coordinate with stakeholders to address priority gaps and improve shared holdings.